Sum-Score Reliability Under Measurement Model Misspecification Is Target-Relative: A Short Note on Omega Simulation Studies

June 28, 2026

Abstract

Simulation studies of coefficient omega under measurement model misspecification usually begin by choosing a population factor model, computing the sum-score reliability implied by that model, fitting alternative models to generated data, and reporting bias relative to the population value. This is a coherent model-relative exercise. It is not, however, a model-free assessment of bias for the composite's reliability. The reason is familiar from the literature on equivalent structural equation models: the observed covariance matrix does not, by itself, identify the substantive latent decomposition. For omega as a composite reliability coefficient, the estimand is the proportion of sum-score variance attributed to a named target factor. Equivalent decompositions can therefore imply the same item covariance matrix while assigning different reliabilities to the same sum score. We use this point to reinterpret recent omega simulations as conditional sensitivity analyses and argue that future simulations should separate estimator behavior from target uncertainty.

Keywords: coefficient omega, sum-score reliability, composite reliability, equivalent models, bifactor model

1 The Problem

Coefficient omega is attractive because it asks a substantively meaningful question: how much of the observed sum score is attributable to the latent variable the score is intended to measure? That question is also what makes omega fragile. The answer depends on which latent variable is treated as the target construct and which parts of the item covariance structure are treated as nuisance, method, local dependence, specific-factor variance, or error.

Bell, Chalmers, and Flora [1] give a useful recent example. Their simulation study is explicitly about reliability of the composite score $X = \mathbf{1}^\mathsf{T} y$. They generate item covariances from four population families: a simple one-factor model, a one-factor model with correlated errors, a bifactor model, and a higher-order model. They then fit one-factor, correlated-error one-factor, and bifactor models and compare the resulting omega estimates with the omega value implied by the population model. In their notation, ω_u is the sum-score reliability with respect to the single factor in a one-factor model; ω_H is the sum-score reliability with respect to the general factor in a bifactor model; and ω_{ho} is the sum-score reliability with respect to the higher-order factor through its indirect effects.

The present note argues that there is a deeper target issue behind such simulation results. Once a population model and target factor are stipulated, the bias calculation is well-defined. But the word "true" supplies not only a covariance matrix. It also supplies a substantive allocation of covariance to the target factor rather than to other latent or residual sources. Under model misspecification, that allocation is often exactly what is in question.

This is not a new problem in structural equation modeling. Equivalent models can fit the same covariance matrix while representing different substantive stories [2, 5]. Fit alone cannot determine which story is the data-generating one. The same difficulty becomes sharper for sum-score reliability because the estimand itself can change across equivalent stories. Reliability is not always a single property of a test; it can depend on what the score is meant to measure [3, 4, 6].

2 Sum-Score Omega as a Target-Factor Quantity

Let y be the vector of item scores and let $X = \mathbf{1}^\mathsf{T} y$ be the sum score. Suppose a model says that a target factor f, scaled to have variance one, contributes loading vector λ to the items:

$$y = \lambda f + u, \qquad \mathrm{Cov}(f, u) = 0, \qquad \mathrm{Var}(y) = \Sigma.$$

The omega target for the sum score, with respect to f, is

$$\omega_f = \frac{\mathrm{Var}(\mathbf{1}^\mathsf{T}\lambda f)}{\mathrm{Var}(\mathbf{1}^\mathsf{T} y)} = \frac{(\mathbf{1}^\mathsf{T}\lambda)^2}{\mathbf{1}^\mathsf{T}\Sigma\mathbf{1}}. \tag{1}$$

This definition uses the full sum-score variance in the denominator. If the model contains correlated residuals, specific factors, or lower-order factors, their contribution remains in $\mathbf{1}^\mathsf{T}\Sigma\mathbf{1}$. What changes from model to model is the numerator: which part of the item covariance is assigned to the named target factor.

Thus, when Bell et al. label a population condition as having reliability .60 or .85, that label is not a property of the covariance matrix alone. It is a property of the covariance matrix together with a particular latent interpretation: a particular λ is declared to be the contribution of the target factor to the items.

3 Equivalent Models Can Change the Sum-Score Target

The issue can be seen with a simple construction. It is not meant as a proposed measurement model. It is only a way to show what is, and is not, identified by the covariance matrix. Let Σ be any positive definite item covariance matrix. For any value r between zero and one, set

$$\lambda_r = \sqrt{r}\frac{\Sigma\mathbf{1}}{\sqrt{\mathbf{1}^\mathsf{T}\Sigma\mathbf{1}}}, \qquad \Psi_r = \Sigma - \lambda_r\lambda_r^\mathsf{T}.$$

Then

$$\Sigma = \lambda_r\lambda_r^\mathsf{T} + \Psi_r$$

is a one-factor model with correlated residuals. The residual covariance Ψ_r is nonnegative definite by Cauchy's inequality, and

$$(\mathbf{1}^\mathsf{T}\lambda_r)^2 = r\,\mathbf{1}^\mathsf{T}\Sigma\mathbf{1}.$$

Equation (1) therefore gives sum-score reliability r with respect to the common factor in that decomposition.

Thus, if correlated residuals are allowed without further restriction, the same observed covariance matrix can be paired with many different sum-score reliabilities. An omega estimator that converges to a fixed function of Σ can therefore be positively biased, unbiased, or negatively biased depending only on which equivalent latent decomposition is called the population model.

4 A Worked Bell-Style Bifactor Example

Consider six standardized items arranged in two three-item clusters, mimicking the bifactor structure used by Bell et al. Let every item load .50 on a general factor and .40 on one of two group factors, with all factors orthogonal and all item residuals independent. The residual variance for each item is

$$1 - .50^2 - .40^2 = .59.$$

The implied item covariance matrix has diagonal entries 1, within-cluster off-diagonal entries

$$.50^2 + .40^2 = .41,$$

and between-cluster off-diagonal entries

$$.50^2 = .25.$$

For the sum score $X = \mathbf{1}^\top y$, the total variance is

$$\mathrm{Var}(X) = 6 + 2\{6(.41) + 9(.25)\} = 15.42.$$

The general factor contributes

$$(\mathbf{1}^\top \lambda_g)^2 = (6 \cdot .50)^2 = 9,$$

so the Bell-style bifactor target is

$$\omega_H = \frac{9}{15.42} = .584.$$

Now hold the item covariance matrix fixed. The same covariance matrix can also be written as a one-factor correlated-error model

$$\Sigma = \lambda_r \lambda_r^\top + \Psi_r$$

with a different target loading vector. If the desired sum-score reliability is $r = .75$, the construction above gives equal loadings

$$\lambda_r = .566789\,\mathbf{1}.$$

The residual covariance matrix then has diagonal entries .678750, within-cluster off-diagonal entries .088750, and between-cluster off-diagonal entries $-.071250$. These residual covariances reproduce exactly the same item covariance matrix, but the sum-score reliability with respect to the common factor is

$$\frac{(6 \cdot .566789)^2}{15.42} = .75.$$

If instead $r = .40$, the equal common-factor loading is .413924 and the residual covariance entries are .828667 on the diagonal, .238667 within clusters, and .078667 between clusters. The same observed covariance matrix now has target-factor sum-score reliability .40.

The point is not that these alternative correlated-error decompositions are substantively preferable to the bifactor model. The point is that they are observationally equivalent at the covariance level and define different sum-score reliability targets. In the bifactor interpretation the population omega is .584; in the two one-factor local-dependence interpretations it is .75 or .40. The denominator, $\mathrm{Var}(X) = 15.42$, has not changed. Only the numerator, the part of the sum-score variance assigned to the target factor, has changed.

5 A Speculative Projection Interpretation

There may still be a useful way to interpret omega estimates from a misspecified one-factor fit. At the population level, fitting a congeneric model to a covariance matrix Σ selects a pseudo-true covariance matrix

$$\Sigma_* = \lambda_* \lambda_*^\mathsf{T} + \Psi_*$$

inside the congeneric model class. The meaning of "closest" depends on the discrepancy: unweighted least squares gives one projection, normal-theory maximum likelihood gives another, and GLS or WLS give still others. The fitted omega can then be viewed as the sum-score reliability of the nearest congeneric approximation,

$$\omega_* = \frac{(\mathbf{1}^\mathsf{T} \lambda_*)^2}{\mathbf{1}^\mathsf{T} \Sigma_* \mathbf{1}}$$

or, if the observed denominator is used, as

$$\omega_*^{obs} = \frac{(\mathbf{1}^\mathsf{T} \lambda_*)^2}{\mathbf{1}^\mathsf{T} \Sigma \mathbf{1}}.$$

This projection interpretation is not the same as an equivalent-model interpretation. The fitted residual

$$\Delta = \Sigma - \Sigma_*$$

is generally not a covariance matrix. For example, under ordinary ULS with free residual variances and an interior solution, the diagonal of Σ_* matches the diagonal of Σ. Then Δ has zero diagonal, so a nonzero Δ cannot be positive semidefinite. It is a discrepancy residual, not an additional source of reliable covariance.

The more promising question is whether the fitted loading vector can be kept while the residual part is merged back into the original covariance:

$$\Sigma = \lambda_* \lambda_*^\mathsf{T} + E_*, \qquad E_* = \Sigma - \lambda_* \lambda_*^\mathsf{T}.$$

If $E_* \succeq 0$, then the misspecified one-factor loading vector defines an exact one-factor model with correlated residuals for the original Σ. In that special case, ω_*^{obs} is not merely the reliability of the nearest congeneric approximation. It is also a legitimate sum-score reliability target for an equivalent local-dependence model. If E_* is not positive semidefinite, the same number is still an algebraic projection index, but it is not literally a variance decomposition of the original covariance matrix.

This suggests a useful diagnostic for misspecification simulations. Alongside the usual bias relative to the declared population target, one can ask whether the fitted one-factor loadings can be embedded in the true covariance as a valid correlated-residual model. Algebraically, this requires

$$\Sigma - \lambda_* \lambda_*^\mathsf{T} \succeq 0,$$

equivalently, for nonsingular Σ,

$$\lambda_*^\mathsf{T} \Sigma^{-1} \lambda_* \leq 1.$$

When the condition holds, the fitted one-factor omega has a possible target-factor interpretation under an equivalent local-dependence model. When it fails, the omega remains interpretable as a projection summary, but not as the reliability of an actual common-factor component in Σ.

6 Re-reading Bell et al.'s Simulation Results

The example clarifies what Bell et al.'s [1] simulations do and do not show. Their population models define clear conditional targets. For a one-factor correlated-error population, the target is the part of sum-score variance due to the one common factor. For a bifactor population, the target is the part due to the general factor. For a higher-order population, the target is the part due to the higher-order factor through its indirect effects.

The limitation is that these are target-relative claims. In a bifactor population,

$$\Sigma = \lambda_g \lambda_g^{\mathsf{T}} + \Lambda_s \Lambda_s^{\mathsf{T}} + \Theta,$$

the Bell target treats $\lambda_g \lambda_g^{\mathsf{T}}$ as construct-relevant general-factor variance and treats $\Lambda_s \Lambda_s^{\mathsf{T}} + \Theta$ as non-target variance for the total score. Marginally, however, $\Lambda_s \Lambda_s^{\mathsf{T}} + \Theta$ is structured residual covariance around the general factor. Another one-factor correlated-residual decomposition can imply the same Σ but assign a different amount of sum-score variance to its common factor. The observed item covariance matrix cannot decide which target is substantively correct.

The higher-order case illustrates an important qualification. A higher-order model and a restricted bifactor model can be related by the Schmid-Leiman transformation [7, 8]. When that transformation preserves the vector of general-factor contributions to the items, omega is preserved too. Equivalent models are not automatically a problem for sum-score reliability. They become a problem when the equivalent representations change the target loading vector in Equation (1).

Thus, simulations of misspecified omega estimates are best read as answering a conditional question:

If this latent target story is true, how does this fitted omega behave?

They do not answer the unconditional question:

How biased is omega for the reliability encoded in the covariance matrix?

There is no such covariance-only sum-score reliability target without further restrictions.

7 Implications

The practical implication is not that omega should be abandoned, or that misspecification simulations are useless. The implication is that simulations should separate two problems that are often blended together.

First, there is an estimator problem. Given a target model, a target factor, and a population covariance matrix, what does a fitted omega estimate in finite samples or asymptotically? Bell et al. address this problem.

Second, there is a target problem. Given the same observed covariance matrix, which part of the sum-score variance should count as construct-relevant reliability? This problem is not solved by Monte Carlo replication, larger sample size, or better fit indices. It requires theory, design information, external validation, or explicit sensitivity analysis over plausible equivalent or near-equivalent decompositions.

Future simulation studies can make this distinction explicit by reporting their results as conditional on a declared target factor, by noting whether the target is invariant across relevant equivalent models, and by including sensitivity conditions that move covariance between general factors, specific factors, and correlated residuals while holding the item covariance matrix fixed or nearly fixed. Such designs would not merely ask whether an estimator is biased under misspecification. They would ask whether the proposed sum-score reliability target itself is stable under the kinds of model uncertainty that applied researchers face.

8 Conclusion

The established lesson from equivalent-model work in SEM is that the same covariance matrix can support different substantive explanations. For model-based composite reliability coefficients, the consequence is stronger: different explanations can define different reliabilities for the same sum score. Bias under misspecification is therefore not a model-free property of coefficient omega. It is a property of an estimator relative to a declared latent target. That framing preserves the value of recent omega simulations while clarifying their scope. They are conditional sensitivity studies, not definitive evidence about bias for a uniquely identified reliability of the observed composite.

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